Activity Diagram

Develop a comprehensive activity diagram for below case study. The diagram should clearly depict all activities, decision points, swimlanes for different actors, transitions, and concurrent processes, aligning with the workflow described in the case study provided below. Include annotations for initial and final nodes, forks, joins, and conditions for transitions wherever applicable.

Autonomous Drone Emergency Delivery Decision Workflow

An emergency response system manages autonomous drone deliveries during disaster situations. When an emergency request is generated by a Requester, the request is submitted to the Central Control System. The Central Control System verifies the authenticity of the request. If the request is found to be invalid, the system logs the request and terminates the process.

If the request is valid, the Central Control System checks drone availability and weather conditions using data from the Monitoring Services. If either the drone is unavailable or the weather conditions are unsafe, the system places the request in a waiting queue and ends the workflow. If conditions are safe, the Central Control System creates a delivery mission and assigns it to an Autonomous Drone. At this stage, two activities occur in parallel: the Central Control System plans the optimal flight path, while the Autonomous Drone prepares the payload and performs system diagnostics.

Once both activities are completed, the Autonomous Drone begins the delivery. During the flight, the drone continuously monitors its battery level and reports status updates to the Central Control System. If the battery level falls below a safe threshold, the drone aborts the mission and returns to base.If the delivery is successfully completed, the Autonomous Drone confirms the delivery to the Central Control System. The system then updates the mission status, notifies the Requester, and terminates the process.

Activity Diagram Evaluation Rubrics (10 Marks)

Criterion	
Understanding of Scenario	2
Initial & Final Nodes	1
Action States	2
Decision & Merge Nodes	2
Fork & Join (Parallelism)	1
Swimlanes / Responsibility Separation	1
UML Notation & Diagram Clarity	1
Total	10 Marks